01 Selecting & Sorting

Start with the basics: pulling data out of the table, removing duplicates, sorting it, and limiting how many rows you get back. The `employees` table below is the data source for every question in this exercise.

▸ TABLE employees

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

01 **Retrieve all columns from the employees table.**

EXPECTED COLUMNS: `id` `first_name` `last_name` `department` `salary` `hire_date` `city`

02 **Find all unique departments.**

EXPECTED COLUMNS: `department`

03 **Retrieve first and last names ordered by salary descending.**

EXPECTED COLUMNS: `first_name` `last_name`

04 **Retrieve the top 3 highest-paid employees.**

EXPECTED COLUMNS: `id` `first_name` `last_name` `salary`

★ REMINDER — SELECT ORDER MATTERS

SQL clauses must appear in this order: `SELECT` → `FROM` → `WHERE` → `ORDER BY` → `LIMIT`. To find the "top 3 highest-paid," combine `ORDER BY salary DESC` with `LIMIT 3`.

02 Filtering with WHERE

Now narrow down your results. The `WHERE` clause lets you keep only the rows that match a condition — combine conditions with **AND**, **OR**, **NOT**, and **IN** for more powerful queries.

↩ **Refer to the `employees` table on the previous page** — it's the same dataset for every question in this exercise.

05 **Find employees in the IT department.**

EXPECTED COLUMNS: `id` `first_name` `last_name` `department`

06 **Find employees in Finance with salary > 60000.**

EXPECTED COLUMNS: `id` `first_name` `last_name` `department` `salary`

07 **Find employees in HR or Marketing.**

EXPECTED COLUMNS: `id` `first_name` `last_name` `department`

08 **Find employees not in IT.**

EXPECTED COLUMNS: `id` `first_name` `last_name` `department`

09 **Find employees in IT, HR, or Finance using IN.**

EXPECTED COLUMNS: `id` `first_name` `last_name` `department`

10 **Find employees in IT with salary > 65000 and city Johannesburg.**

EXPECTED COLUMNS: `id` `first_name` `last_name` `department` `salary` `city`

★ BEFORE YOU SUBMIT — CHECK YOUR WORK

- Did you write all **10** SQL queries?
- Does each output use the exact column names listed under *Expected columns*?
- Did you draw a clear output table for every query — even single-row results?
- Did you wrap text values in single quotes? E.g. `WHERE department = 'IT'`.
- Are your queries readable — keywords in CAPS, one clause per line where helpful?